

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #04 — Campus & Academic Operations: Student Project Portfolio & Showcase App

Annual project expos require a centralized showcase where student teams submit multimedia artifacts, judges grade projects using digital rubrics, and the public participates in voting with strict anti-sybil protections. This document contains the three required deliverables: (1) the complete Functional & Non-Functional Requirements table, (2) the UML Use-Case Diagram, and (3) the detailed Use-Case Flow Specification for one core use case.

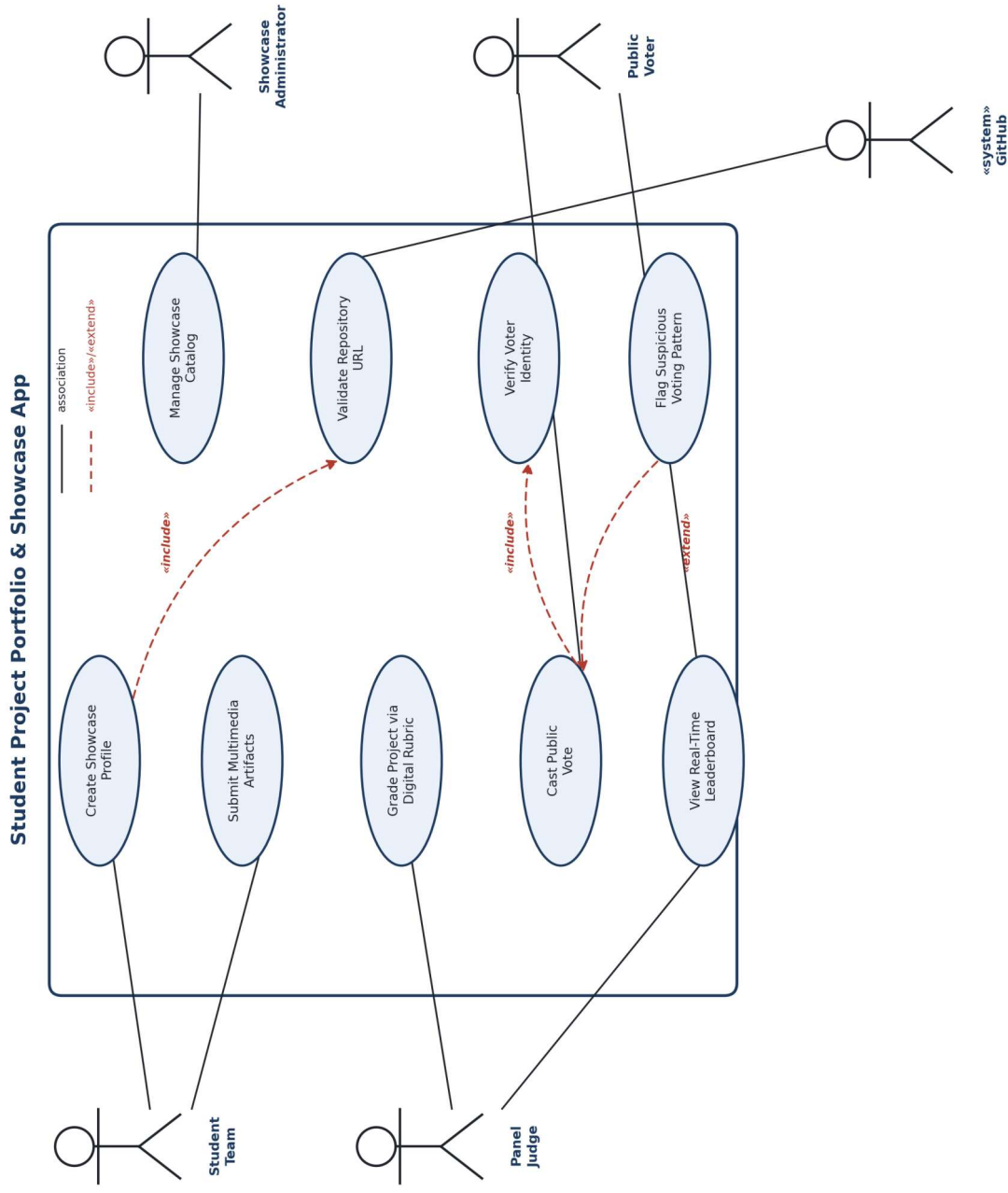
1. Requirements Table

Functional Requirements FR-001 – FR-005 and Non-Functional Requirements NFR-001 – NFR-002.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional – Profile Management	The system shall enable student teams to create showcase profiles containing abstract summaries, demo video links, and verified GitHub repository URLs.	High	Pass: Profile is published to the showcase catalog upon validation of the repository link. Fail: An invalid URL format is accepted.	Establishes the canonical, verifiable project record that judges and the public rely on for evaluation and discovery.
FR-002	Functional – Digital Rubric Grading	The system shall allow panel judges to score each showcase profile against a configurable, weighted rubric (e.g., innovation, technical depth, presentation) and submit per-criterion scores.	High	Pass: Submitted rubric scores are time-stamped and locked against the judge's identity. Fail: A score outside the defined 0–10 range per criterion is accepted.	Guarantees consistent, auditable, and comparable evaluation of every team across all judges.
FR-003	Functional – Public Voting & Anti-Sybil	The system shall let members of the public cast exactly one vote per showcase profile after completing an anti-sybil verification step (OTP-verified contact plus device fingerprint check).	High	Pass: A second vote attempt from the same verified identity/device is rejected as duplicate. Fail: More than one vote from the same verified voter is recorded.	Protects the integrity and public trust of the voting results against bot- and duplicate-driven manipulation.
FR-004	Functional – Multimedia Artifact Management	The system shall allow student teams to upload and manage demo videos and poster images for their profile, validating file type and size before accepting the upload.	Medium	Pass: Only approved formats (MP4/MOV ≤200MB video, PNG/JPG ≤10MB image) are stored. Fail: An unsupported file type or oversized file is accepted.	Ensures showcase media is playable and consistently sized for reliable display to judges and visitors.
FR-005	Functional – Real-Time Leaderboard	The system shall compute and display a real-time leaderboard combining normalized judge rubric scores and public vote counts using a configurable weighting formula.	High	Pass: Leaderboard position updates within 300 ms of a new vote or score submission. Fail: The leaderboard shows stale rankings after a new score/vote is recorded.	Gives teams, judges, and attendees immediate, trustworthy visibility into standings during the live expo.
NFR-001	Performance & Security	The public voting and leaderboard calculations shall update in real-time with an API response latency under 300 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Expo-day traffic spikes sharply during finalist reveals; slow or insecure voting would undermine credibility of the results.
NFR-002	Reliability & Availability	The system shall maintain at least 99.9% uptime during the official judging and voting window, with automatic recovery of in-progress rubric scores/votes (no data loss) after a dropped connection.	High	Pass: Simulated disconnect tests show in-progress submissions auto-save within 5 seconds and are recoverable on reconnect. Fail: Any submitted score or vote is lost after a simulated disconnect.	A single-day, high-stakes event cannot tolerate downtime or lost submissions without damaging the competition's fairness and reputation.

2. UML Use-Case Diagram

Primary actors **Student Team** and **Panel Judge**, plus **Public Voter** and **Showcase Administrator**, interact with the system boundary. The external system actor **GitHub** supports repository validation. The diagram includes one «include» relationship (Create Showcase Profile → Validate Repository URL; Cast Public Vote → Verify Voter Identity) and one «extend» relationship (Flag Suspicious Voting Pattern extends Cast Public Vote).



3. Use-Case Flow Specification

Use Case Name	Cast Public Vote
Use Case ID	UC-05
Primary Actor	Public Voter
Secondary Actors	Anti-Sybil Verification Service (included), System (leaderboard update)
Related Requirements	FR-003, NFR-001
Trigger	A public voter selects "Vote for this project" on a showcase profile page during an active voting window.

Preconditions

1. The showcase catalog is published and the official public-voting window is open.
2. The target project profile has passed repository validation (FR-001) and appears in the public catalog.
3. The voter has network access to the public voting portal (web or mobile).

Postconditions — Success

Exactly one vote is recorded for the selected profile, linked to the voter's hashed verified-identity/device fingerprint (no personally identifying data retained), and the real-time leaderboard reflects the updated count within 300 ms (NFR-001).

Postconditions — Failure

No vote is recorded and system state is unchanged, apart from an audit-log entry noting the rejected attempt.

Main Success Scenario

1. Voter opens a project's showcase profile and selects "Vote for this project."
2. System invokes the included use case **Verify Voter Identity**: voter supplies a phone number or email address.
3. System sends a one-time password (OTP); voter enters it while the system silently captures a device fingerprint.
4. System checks the verified identity/device fingerprint against the anti-sybil vote ledger for this event (*extension point: AnomalyCheck*).
5. System confirms no prior vote exists from this identity for this profile/event.
6. System records the vote against the hashed voter identity, enforcing the one-vote-per-voter rule, without storing personally identifying details.
7. System updates the real-time leaderboard and displays a confirmation, "Your vote has been counted," to the voter.

Alternate Flow A1 — Duplicate Vote Detected

Branches at Step 5. Trigger: the anti-sybil ledger already holds a recorded vote for this verified identity/device against this profile or event.

- 5a.1 System rejects the vote attempt and displays "You have already voted for this project."
- 5a.2 System logs the duplicate attempt (identity hash, timestamp, device fingerprint) for audit purposes.
- 5a.3 Use case ends; the leaderboard remains unchanged.

Extension — Flag Suspicious Voting Pattern («extend»)

At extension point *AnomalyCheck* (after Step 4), if the system detects an anomalous cluster of votes (e.g., many verified identities sharing a device fingerprint or IP subnet in a short window), it extends the base flow to flag the pattern for administrator review without blocking individual legitimate voters.

Special Requirements

Voting and OTP delivery must meet the NFR-001 latency target (<300 ms API response) even under simulated peak expo-day load; voter contact details used for OTP must not be exposed in any public-facing leaderboard or export.